

Homework 2

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Convention 0.1. We keep with the book's notation and write df for the differential of f , rather than f_* .

Remark 0.2. In the previous homework, I showed maps are smooth if and only if they are smooth componentwise. This result will be continuously referenced by referencing this remark.

Problem 3.1

Problem 1. Let M, N be smooth manifolds with or without boundary, and $F : M \rightarrow N$ a smooth map. Show that $dF_p : T_p M \rightarrow T_{F(p)} N$ is the zero map for each $p \in M$ if and only if F is constant on each component of M .

Proof. ($\Leftarrow$) Let $F : M \rightarrow N$ be constant on all connected components of M . We start by assuming M is connected, and use this result to generalize. Let $v \in T_p M$, and $f \in C^\infty(N)$. It suffices to show $dF(v)_p(f) = v(f \circ F) = 0$. This indeed follows, since F is the constant map, and thus so is $f \circ F$. Derivations of constant maps are 0, so the statement follows.

For the more general case, let $p \in U \subseteq M$ such that U is a connected component of M , and note the isomorphism $T_p M \cong T_p U$ which follows by reasoning explained later. By hypothesis, F is constant on U , meaning any composite $f \circ F$ restricted along U is constant. The derivation $d(F|_U)_p : T_p U \rightarrow T_{F(p)} N$ is then necessarily zero, since derivations of constant maps are 0. By the isomorphism $T_p M \cong T_p U$, this suffices for the arbitrary case, since $dF_p(v) = 0$ in $T_p M$ if and only if $dF_p(v) = 0$ in $T_p U$.

($\Rightarrow$) Let $dF_p : T_p M \rightarrow T_{F(p)} N$ be zero for all $p \in M$. We start by assuming M is connected, then generalize similarly to the ($\Leftarrow$) case. Since path-connectedness is a topological invariant, and since $\mathbb{R}^n$ is path connected, the locally Euclidean property of manifolds implies they are locally path-connected. This also follows for manifolds with boundary since $\mathbb{H}^n$ is path-connected. By Lee A.43, locally path connected spaces are connected if and only if they are path connected, and thus manifolds are path connected. Note that connected components are open in locally path-connected spaces, hence components of manifolds form open submanifolds. This justifies the isomorphisms $T_p M \cong T_p U$ for U a connected component.

It suffices to show $F(x) = F(y)$ for any $x, y \in M$. By path-connectedness, let $\gamma : I \rightarrow M$ be a smooth path with $\gamma(0) = x, \gamma(1) = y$. Note that $F \circ \gamma : I \rightarrow N$ gives a path from $F(x)$ to $F(y)$. By assumption this map is smooth, so it has a velocity vector

$$\frac{d}{dt}(F \circ \gamma)(t_0) = dF \left(\left. \frac{d\gamma}{dt} \right|_{t_0} \right) = dF_{\gamma(t_0)} \left(\left. \frac{d\gamma}{dt} \right|_{t_0} \right).$$

By assumption, $dF_{\gamma(t_0)} = 0$ since it is 0 at each point p , meaning

$$\frac{d}{dt}(F \circ \gamma)(t) = 0.$$

A path with everywhere zero velocity is necessarily constant, meaning $F \circ \gamma$ is constant. Thus, $F(x) = F(y)$, and F is the constant map.

Now take M to be an arbitrary manifold. Let $p \in M$, and choose a connected component $U \subseteq M$ containing p . Since U is connected, and since $d(F|U)_p : T_p U \rightarrow T_{F(p)} N$ is zero by the isomorphism $T_p M \cong T_p U$, we necessarily have that F is constant on U by the previous case. Therefore, F is constant on each component. ■

Remark 1.1. I felt my rigor when justifying that $d(F|U)_p = 0$ if and only if $dF = 0$ was lacking and messy, so I have added some further commentary. The restriction $F|U$ is simply the composite $F \circ i$, for $i : U \hookrightarrow M$ the inclusion. Since this is smooth, $d(F|U)_p = d(F \circ i)_p = dF_p \circ di_p$. We see that if $dF_p = 0$, then the whole composite is necessarily 0 by properties of the 0 morphism.

Conversely, if $d(F|U)_p = 0$, then $dF_p \circ di_p = 0$. However, di_p is an isomorphism, meaning it is nonzero whenever $T_p U$ is nonzero. It follows in this case that dF_p is necessarily 0, since no other map composed with an isomorphism is 0 since units are not zero-divisors by ring theory. Otherwise, if $T_p U \cong 0$, then $T_p M \cong T_p U$ implies $T_p M \cong 0$, and dF_p is necessarily 0 since the only map $0 \rightarrow T_{F(p)} N$ is the zero morphism. (Or I could have just done $0 = 0 \circ di_p^{-1} = dF_p \circ di_p \circ di_p^{-1} = dF_p$)

Problem 3.2

Problem 2. Let $\{M_i\}_{i=1}^k$ be smooth manifolds, and let $\pi_j : \prod_i M_i \rightarrow M_j$ be the natural projection. Prove that for any point $p = p_i \in \prod_i M_i$, the map

$$\alpha : T_p \prod_{1 \leq i \leq k} M_i \rightarrow \bigoplus_{1 \leq i \leq k} T_{p_i} M_i$$

given by

$$\alpha(v) = (d(\pi_1)_p(v), \dots, d(\pi_k)_p(v))$$

is an isomorphism. The same is true if one of the spaces M_i has boundary.

Proof. We will never deal with the global differential in this problem, so we drop the subscript when referring to differentials for notational convenience, and also because I accidentally forgot the subscripts when writing it up and don't want to fix it.

Because the product and coproduct in $\mathbb{R}\text{-Vect}$ coincide, we start by defining the map

$$\iota_i : M_i \rightarrow \prod_{i=1}^k M_i, \quad q \mapsto (p^1, p^2, \dots, p^{i-1}, q, p^{i+1}, \dots, p^n).$$

We note that ι_i maps $p^i \mapsto p$, so the differential gives a map $d\iota_i : T_{p^i} M_i \rightarrow T_p \prod_i M_i$. Universality assembles these into a map

$$\begin{array}{ccc} T_{p^i} M_i & & \\ \downarrow & \searrow d\iota_i & \\ \bigoplus_{i=1}^k T_{p^i} M_i & \xrightarrow{\beta} & T_p \prod_{i=1}^k M_i \end{array}$$

defined by

$$\beta(v)(f) = \sum_{i=1}^k d\iota_i(v)(f).$$

It suffices of course to show that ι_i is C^∞ , otherwise its differential need not meaningfully exist, and also that α and β are indeed inverses.

By remark 0.2, maps are smooth if and only if they are smooth componentwise. The $j \neq i$ th components map $x \mapsto p^j$, and i th component maps $x \mapsto x$, which is simply the identity. It was shown in the last homework that the identity is smooth, so we need only show the constant map is smooth. This is probably obvious enough that a proof need not be included, but I want extra practice with smoothness proofs. We do this for arbitrary manifolds M, N . Fix some $p \in N$, and let $f : M \rightarrow N$ be the constant map at p . Let $x \in M$, and choose a chart (U, φ) of x . Choose any chart (V, ψ) of $p = f(x)$, and $f(U) = \{p\} \subseteq V$. Note also that

$$(\psi \circ f \circ \varphi^{-1})(\hat{x}) = (\psi \circ f)(x) = \psi(p).$$

Thus, $\psi \circ f \circ \varphi^{-1}$ is the constant map at $\psi(p)$, which is smooth by calc 3 knowledge. This shows that ι_i is smooth, and thus its differential is well-defined. Therefore, β is well-defined by universality.

It suffices at this point to show α and β are inverses. Let $v \in T_p \prod_i M_i$, and let $f : \prod_i M_i \rightarrow \mathbb{R}$ be smooth. We first note that

$$(\beta\alpha)(v) = \beta(d\pi_1(v), \dots, d\pi_k(v)) = \sum_{i=1}^k d\iota_i d\pi_i(v) = \sum_{i=1}^k d(\iota_i \circ \pi_i)(v).$$

By linearity of derivations, mapping under f gives

$$(\beta\alpha)(v)(f) = \left(\sum_{i=1}^k d(\iota_i \circ \pi_i)(v) \right) (f) = \sum_{i=1}^k d(\iota_i \circ \pi_i)(v)(f) = \sum_{i=1}^k v(f \circ \iota_i \circ \pi_i).$$

Since v is linear, it will be useful to factor it out of the sum as

$$\sum_{i=1}^k v(f \circ \iota_i \circ \pi_i) = v \left(\sum_{i=1}^k f \circ \iota_i \circ \pi_i \right).$$

Let $x \in \prod_i M_i$. Chasing x along the i th component of the sum gives

$$x \xrightarrow{\pi_i} x^i \xrightarrow{\iota_i} (p^1, \dots, p^{i-1}, x^i, p^{i+1}, \dots, p^n) \xrightarrow{f} f(p^1, \dots, x^i, \dots, p^n).$$

Summing over k yields

$$\left(\sum_{i=1}^k f \circ \iota_i \circ \pi_i \right) (x) = \sum_{i=1}^k f(p^1, \dots, x^i, \dots, p^n).$$

Since the action of v is simply given by $v^i \partial_i$, it suffices to show f and $\sum_i f \circ \iota_i \circ \pi_i$ have the same partials at p .

Note that since $f \circ \iota_i \circ \pi_i$ is constant at all x^j for $j \neq i$, and is simply f on x^i , we see that its partial with respect to x^j is 0 whenever $j \neq i$, and is simply $\partial_i f$ when $j = i$. It follows that

$$\left. \frac{\partial}{\partial x^j} (f \circ \iota_i \circ \pi_i) \right|_p = \delta_{ij} \left. \frac{\partial f}{\partial x^j} \right|_p$$

for δ_{ij} the Kronecker delta. We thus have

$$\left. \frac{\partial}{\partial x^j} \left(\sum_{i=1}^k f \circ \iota_i \circ \pi_i \right) \right|_p = \sum_{i=1}^k \left. \frac{\partial}{\partial x^j} (f \circ \iota_i \circ \pi_i) \right|_p = \sum_{i=1}^k \delta_{ij} \left. \frac{\partial f}{\partial x^j} \right|_p = \left. \frac{\partial f}{\partial x^j} \right|_p.$$

Therefore, the partials of f and $\sum_i f \circ \iota_i \circ \pi_i$ agree, meaning v acts on them the same way. This suffices to prove this direction.

The converse direction is more simple. Let $v = (v^1, \dots, v^n) \in \bigoplus_{i=1}^k T_{p^i} M_i$. We pass to Einstein notation and write v^i for v to simplify the notation. Then

$$(\alpha\beta)(v) = \alpha(d\iota_i(v^i)) = (d\pi_1 d\iota_i(v^i), \dots, d\pi_k d\iota_i(v^i)) = (d(\pi_1 \iota_i)(v^i), \dots, d(\pi_k \iota_i)(v^i)),$$

where the last step follows by linearity of differentials. We now stop using Einstein notation. It suffices to show that $\sum_i d(\pi_j \iota_i)(v^i) = v^j$ for some fixed j . Let $f : M_j \rightarrow \mathbb{R}$ be smooth. Then

$$\sum_{i=1}^k d(\pi_j \iota_i)(v^i)(f) = \sum_{i=1}^k v^i(f \circ \pi_j \circ \iota_i).$$

It suffices to show that for each i ,

$$v^i(f \circ \pi_j \circ \iota_i) = \sum_{j=1}^k \delta_{ij} v^j(f) = v^i(f),$$

for δ_{ij} the Kronecker delta. Indeed, when $i \neq j$, the map $f \circ \pi_j \circ \iota_i$ is the constant map at p^j , which is seen by chasing elements:

$$M_i \ni x \xrightarrow{\iota_i} (p^1, \dots, x, \dots, p^n) \xrightarrow{\pi_j} p^j \xrightarrow{f} f(p^j).$$

Derivations of constant functions are zero, so $v^i(f \circ \pi_j \circ \iota_i) = 0$. Conversely, when $i = j$, then

$$M_j \ni x \xrightarrow{\iota_j} (p^1, \dots, x, \dots, p^n) \xrightarrow{\pi_j} x \xrightarrow{f} f(x).$$

Thus, $f \circ \pi_j \circ \iota_j = f$, and therefore $v^j(f \circ \pi_j \circ \iota_j) = v^j(f)$. We then have

$$(\alpha\beta)(v)(f) = \sum_{i=1}^k (\delta_{i1} v^i(f), \dots, \delta_{ik} v^i(f)) = (v^1(f), \dots, v^k(f)) = v^i(f).$$

This concludes the proof. ■

Problem 3.3

Problem 3. Show that if M, N are smooth manifolds, then there is a diffeomorphism $T(M \times N) \cong T(M) \times T(N)$.

Proof. First, we construct an isomorphism of sets. We then show the map raises to a diffeomorphism. From problem 2, we have that $T_p(M \times N) \cong T_{p^1}(M) \oplus T_{p^2}(N) \cong T_{p^1}(M) \times T_{p^2}(N)$. This already gets us halfway there:

$$T(M \times N) := \coprod_{p \in M \times N} T_p(M \times N) \cong \coprod_{p \in M \times N} (T_{p^1} M \oplus T_{p^2} N) \cong \coprod_{p \in M \times N} (T_{p^1} M \times T_{p^2} N).$$

We could always invoke $- \times Z \dashv \text{Set}(Z, -)$ to complete the proof of the bijection, but an explicit construction of the isomorphism will be useful. An element of the coproduct over $T_{p^1} M \times T_{p^2} N$ is a triple (p, v_p, w_p) , where $p = (p^1, p^2)$, $v_p \in T_{p^1} M$, and $w_p \in T_{p^2} N$. An element of

$$TM \times TN := \left(\coprod_{p \in M} T_p M \right) \times \left(\coprod_{q \in N} T_q N \right)$$

is a tuple (p, v_p, q, w_q) with $v_p \in T_p M$ and $w_q \in T_q N$. We then define the map $(p, v_p, w_p) \mapsto (p^1, v_p, p^2, w_p)$. This map is manifestly injective. Surjectivity is also immediate, since given any $(p, v_p, q, w_q) \in TM \times TN$, we have that $(p, q) \in M \times N$, meaning $((p, q), v_p, w_q) \mapsto (p, v_p, q, w_q)$. Therefore, we have an isomorphism. In particular, this isomorphism is the *identity* on vectors.

We now promote the isomorphism of sets to a diffeomorphism. By construction of the isomorphism $T_p(M \times N) \cong T_{p^1} M \times T_{p^2} N$, we see that the map we have constructed may be described as

$$(p, v_p, w_p) \mapsto (p^1, d(\pi_M)_p(v_p), p^2, d(\pi_N)_p(w_p)).$$

The maps $p \mapsto p^1$ and $p \mapsto p^2$ are simply the projections $M \times N \rightarrow M$ and $M \times N \rightarrow N$, which we proved in the previous homework are smooth. It suffices to show the other two components are smooth, meaning we must show the map $(v, w) \mapsto (d(\pi_M)_p(v), d(\pi_N)(w))$ is smooth for all p . Note that assembling each $d(\pi_M)_p$ along the coproduct yields the global differential, giving us the map

$$d(\pi_M) \times d(\pi_N) : (v, w) \mapsto (d(\pi_M)(v), d(\pi_N)(w)).$$

Since the global differential is smooth, the map $T(M \times N) \rightarrow TM \times TN$ is indeed smooth by remark 0.2.

Now we show the inverse is smooth. The inverse to this map is awful and annoying. We denote the inverse by F because in retrospect, not naming these maps immediately was a mistake. Given $v \in T_p M$ and $w \in T_q N$, recall the inverse is given by

$$(v, w) \mapsto d(\iota_M)_p(v) + d(\iota_N)_q(w).$$

Here, the inclusions ι_M include $T_p M$ into $T_{(p,q)}(M \times N)$ by mapping along $T_p M \times \{q\}$, and the same for N . Since addition is smooth, we would like to assemble the differentials along the coproduct into the global differentials $d(\iota_M)(v) + d(\iota_N)(w)$. It is not immediately clear that we can do this, so we appeal to a short diagram chase.

The map $F : TM \times TN$ arises by constructing the maps $F_{(p,q)} : T_p M \times T_q N \rightarrow T_{(p,q)}(M \times N)$ of problem 2:

$$(v, w) \mapsto d(\iota_M)_p(v) + d(\iota_N)_q(w);$$

and then assembling along the coproduct as indicated by the diagram

$$\begin{array}{ccc} T_p M \times T_q N & \xrightarrow{F_{(p,q)}} & T_{(p,q)}(M \times N) \\ \downarrow i \times i & & \downarrow \\ TM \times TN & \xrightarrow{F} & T(M \times N) \end{array}$$

Note that $F_{(p,q)}$ factors along $T_{(p,q)}(M \times N) \times T_{(p,q)}(M \times N)$ as

$$(v, w) \xrightarrow{d(\iota_M)_p \times d(\iota_N)_q} (d(\iota_M)_p(v), d(\iota_N)_q(w)) \xrightarrow{+} d(\iota_M)_p(v) + d(\iota_N)_q(w).$$

Factoring the coproduct diagram along this decomposition yields

$$\begin{array}{ccccc} T_p M \times T_q N & \xrightarrow{d(\iota_M)_p \times d(\iota_N)_q} & T_{(p,q)}(M \times N)^2 & \xrightarrow{+} & T_{(p,q)}(M \times N) \\ \downarrow i \times i & & \downarrow & & \downarrow \\ TM \times TN & \xrightarrow{d(\iota_M) \times d(\iota_N)} & T(M \times N)^2 & \xrightarrow{+} & T(M \times N) \end{array}$$

Note here that each $T_{(p,q)}(M \times N) \times T_{(p,q)}(M \times N)$ includes into $T(M \times N) \times T(M \times N)$ by taking the product of the inclusions, which is smooth by remark 0.2. We then very clearly observe that the lower arrow $TM \times TN \rightarrow T(M \times N) \times T(M \times N)$ is simply the product of the coproducts along $d(\iota_M)_p$ and $d(\iota_N)_q$, which are precisely the global differentials. Thus, F factors as $+\circ(d(\iota_M) \times d(\iota_N))$. Since global differentials are smooth, products of smooth maps are smooth (remark 0.2), and addition is smooth, F is necessarily smooth. We obtain that the map is a diffeomorphism, as required. ■

Problem 3.4

Problem 4. *Show there is a diffeomorphism $TS^1 \cong S^1 \times \mathbb{R}$.*

Proof. It will be most convenient to view $S^1 \cong \mathbb{R}/\mathbb{Z}$ as a quotient. Consider the vector field $\tilde{V} : \mathbb{R} \rightarrow T\mathbb{R}$ given by $t \mapsto \frac{d}{dx}|_t$. This descends along the quotient $\pi : \mathbb{R} \rightarrow S^1$ to a map $V : S^1 \rightarrow TS^1$ given by

$$[t] \mapsto d(\pi) \left(\frac{d}{dx} \Big|_t \right) = d(\pi)_{[t]} \left(\frac{d}{dx} \Big|_t \right).$$

It might not be clear that this is well-defined, since we evaluate the derivative at a representative of $[t]$. However, since precomposition by π identifies all elements of the class of t , it is indeed well-defined. Moreover, clearly V is a section of the projection $TS^1 \rightarrow S^1$, since the image of $[t]$ is an element of the tangent space $T_{[t]}S^1$. We would like to now show V is smooth.

Since the global differential $d(\pi)$ is smooth, and since $\pi : \mathbb{R} \rightarrow S^1$ is (clearly) smooth, it suffices to show $\tilde{V}$ is smooth, since $V[t] = (d(\pi) \circ \tilde{V})(t)$. Since $\mathbb{R}$ admits a global chart $(\mathbb{R}, 1_{\mathbb{R}})$ for $1_{\mathbb{R}}$ the identity, the global chart on $T\mathbb{R}$ can be described as

$$\left(t, c \frac{d}{dx} \Big|_t \right) \mapsto (t, c).$$

Precomposing by $\tilde{V}$ gives

$$t \mapsto \left(t, 1 \cdot \frac{d}{dx} \Big|_t \right) \mapsto (t, 1).$$

It follows that $\tilde{V}$ composed with the chart is smooth, since one component is the identity $t \mapsto t$, and the other is the constant map at 1, both of which are smooth.

Note that since $\tilde{V}$ was nowhere vanishing, V necessarily is also nowhere vanishing. Thus, $V[t]$ is a nonzero vector in the tangent space $T_{[t]}S^1$ for all $[t] \in S^1$. Since this is a 1-dimensional vector space, any nonzero vector forms a basis, meaning for all $v \in T_{[t]}S^1$, there is a unique $c \in \mathbb{R}$ with $cV[t] = v$. By uniqueness of c , we can construct a well-defined map

$$F : TS^1 \rightarrow S^1 \times \mathbb{R} \quad ([t], v) \mapsto ([t], c),$$

with notation as before. We automatically have that each $T_{[t]}S^1 \rightarrow \mathbb{R}$ is an isomorphism of sets by uniqueness of basis representations, meaning each $\{[t]\} \times T_{[t]}S^1 \rightarrow \mathbb{R}$ is an isomorphism. These assemble along the coproduct to a bijection $TS^1 \cong S^1 \times \mathbb{R}$. It suffices to show the map is smooth, and the evident inverse

$$([t], c) \mapsto ([t], cV[t])$$

is smooth.

Clearly the first component is smooth since it is the identity on S^1 , so it suffices by remark 0.2 to prove the second component. The inverse direction $c \mapsto cV[t]$ is smooth because scalar multiplication is smooth and V is smooth. The converse direction follows since V is smooth, so $v \mapsto c$ is smooth. ■

Problem 3.5

Problem 5. Let $S^1 \subseteq \mathbb{R}^2$ be the 1-sphere, and let $K \subseteq \mathbb{R}^2$ be the boundary of the square $K = \partial([-1, 1] \times [-1, 1])$. Show there is a homeomorphism $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ which restricts along S^1 to give $F(S^1) = K$, but there is no diffeomorphism with this property.

Proof. First we construct the homeomorphism. The explicit construction is quite messy, so we explain how it works before working through it. Each $p \in S^1$ has an angle θ from the x -axis (any line works, but the x -axis is convenient because $p = \exp(i\theta)$). This angle θ forms a line through 0 and p . We push p away from the origin along this line, until one of its cartesian coordinates attains a magnitude of 1, and its other coordinate has lower magnitude. The point is then on the square, and this construction is obviously a homeomorphism. This extends to $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ via polar coordinates.

Each (x, y) can be written as $(R \cos \theta, R \sin \theta)$ for some $\theta \in [0, 2\pi)$ and $R > 0$. Consider first some (x, y) on the square, meaning that

$$\max(|R \cos \theta|, |R \sin \theta|) = 1.$$

We see that

$$R(\theta) = \frac{1}{\max(|\cos \theta|, |\sin \theta|)}.$$

Since the 1 is simply the magnitude of $(\cos \theta, \sin \theta)$, we might hope that we can define a map $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$(x, y) \mapsto \frac{\sqrt{x^2 + y^2}}{\max(|x|, |y|)}(x, y).$$

Of course, at 0 we map $(0, 0) \mapsto (0, 0)$. This consideration means that the denominator is never zero, and in this case division is continuous. Moreover, square roots, products, and exponents are continuous, so the numerator is continuous. It suffices to show the denominator is continuous when $(x, y) \neq 0$, since it should be clear that the map is indeed continuous at 0.

Indeed, the max function (with absolute values) can be thought of as just the projection onto x , composed with $|-|$, whenever $|x| \geq |y|$, and similarly for y . Clearly these two projections agree on the boundary, and are both continuous, so their union is the max function, rendering it continuous. This can be rigorized further by constructing the relevant pushout square.

The map is thus continuous, and by construction we have $F(S^1) = K$. It is also clearly a bijection, since it has an inverse

$$(x, y) \mapsto \frac{\max(|x|, |y|)}{\sqrt{x^2 + y^2}}(x, y),$$

again with the special consideration that $(0, 0) \mapsto (0, 0)$. This is continuous by the exact same argument, since all its component functions are continuous.

Now to show there is no diffeomorphism with this property. Intuitively, K should not be differentiable at a corner, so we will focus on the point $(1, 1) \in K$. Let $F : \mathbb{R}^2 \cong \mathbb{R}^2$ be a diffeomorphism, and suppose for contradiction that $F(S^1) = K$. Since S^1 is a smooth submanifold of $\mathbb{R}^2$, the restriction $F|_{S^1}$ must be diffeomorphic to $F(S^1) = K$. It suffices to show that K is not a smooth manifold.

Consider the point $(1, 1) \in K$, and suppose there is a chart (U, φ) for $(1, 1)$. Let $I := \varphi(U) \subseteq \mathbb{R}$ be an open interval, and let $\gamma : I \rightarrow U$ be the inverse of φ . In particular, γ is a path in K , and for it to be smooth, its tangent vector must exist. Without loss of generality, take $\gamma(0) = (1, 1)$. For $t < 0$, take $\gamma(t) = (x(t), 1)$, and for $t > 0$, take $\gamma(t) = (1, y(t))$. The tangent vectors to $\gamma(t < 0)$

and $\gamma(t > 0)$ are thus $(x'(t), 0)$ and $(0, y'(t))$, respectively. It follows that $x'(t) = y'(t) = 0$, and by continuity the same is true at $(1, 1)$. Thus, γ has everywhere-vanishing velocity, implying that it is a constant map. Since singletons are not open in $\mathbb{R}^2$, and by extension K , we have that γ cannot be a bijection, and thus not a diffeomorphism, a contradiction. ■

Problem 3.6

Problem 6. View the 3-sphere as $S^3 \subseteq \mathbb{C}^2$ by the usual identification $\mathbb{C}^2 \cong \mathbb{R}^4$. For each $z^i \in S^3$, define $\gamma_z : \mathbb{R} \rightarrow S^3$ by $\gamma_z(t) = (e^{it}z^1, e^{it}z^2)$. Show γ_z is smooth with everywhere nonzero velocity.

Proof. It will be most convenient to view S^3 as a subspace of $\mathbb{R}^4$, so note that since $(z^1, z^2) \in S^3$, we have $0 \leq |z_1|, |z_2| \leq 1$, meaning we may write

$$z^1 = r_1 e^{i\theta_1}, \quad z^2 = r_2 e^{i\theta_2}$$

for $0 \leq r_1, r_2 \leq 1$. We then have

$$\gamma_z(t) = (r_1 e^{i(\theta_1+t)}, r_2 e^{i(\theta_2+t)}).$$

By Euler's formula, identifying under $\mathbb{C}^2 \cong \mathbb{R}^4$ gives

$$\gamma_z(t) = (r_1 \cos(\theta_1 + t), r_1 \sin(\theta_1 + t), r_2 \cos(\theta_2 + t), r_2 \sin(\theta_2 + t)).$$

Smoothness now follows fairly simply. The charts $(U_i^\pm, \varphi_i^\pm)$ simply drop a component, so the composites $\varphi_i^\pm \circ \gamma_z$ are simply three of the four components of the path. It thus suffices to show all four are smooth in the usual calc 3 sense. This indeed follows since trig functions, addition, and scalar multiplication are all smooth, and composites of smooth maps are smooth. The map is thus componentwise smooth, and appealing to remark 0.2 yields that γ_z is a smooth path.

Now we show the velocity is everywhere nonvanishing. Given a chart (U, φ) for $\gamma_z(t_0) \in S^3$, we have the coordinate representation

$$\gamma'_z(t_0) = \frac{d\gamma_z^i}{dt}(t_0) \frac{\partial}{\partial x^i} \Big|_{\gamma_z(t_0)},$$

where γ^i is the i th coordinate function $\varphi^i \circ \gamma$. Since partials $\partial_i|_{\gamma(t_0)}$ are never identically zero, it suffices to show the derivatives of $\varphi^i \circ \gamma$ are everywhere nonzero.

Since $z \in S^3$, at least one of z^1, z^2 is nonzero. Without loss of generality, take $z^1 \neq 0$. It follows that $r_1 \neq 0$. It's possible that one of the first two components will be zero when $\theta_1 + t = n\pi/2$, however both cannot be zero simultaneously simply by knowledge of (co)sine. We can separate the problem into two cases: one where $r_1 \cos(\theta_1 + t)$ is nonzero, and one where $r_1 \sin(\theta_1 + t)$ is nonzero. The logic is exactly the same for both, so we solve only the first case.

Since $r_1 \cos(\theta_1 + t)$ is nonzero, it falls in some chart $(U_1^\pm, \varphi_1)$. The velocity then takes the form

$$\gamma'_z(t_0) = \frac{d(\varphi_1^\pm \circ \gamma_z)}{dt}(t_0) \frac{\partial}{\partial x^i} \Big|_{\gamma_z(t_0)}.$$

Since $\varphi_1^\pm$ drops the first component, we see that

$$(\varphi_1^\pm \circ \gamma_z)(t_0) = (r_1 \sin(\theta_1 + t), r_2 \cos(\theta_2 + t), r_2 \sin(\theta_2 + t)).$$

The derivatives of this map are

$$\frac{d}{dt}(\varphi_1^\pm \circ \gamma_z) = (r_1 \cos(\theta_1 + t), -r_2 \sin(\theta_2 + t), r_2 \cos(\theta_2 + t)).$$

Of course, we do not know that the final two components are nonzero, so we must show the first component is nonzero while t_0 is such that $\gamma_z(t_0) \in U_1^\pm$. Indeed, note that the first component of $\gamma_z(t)$ was $r_1 \cos(\theta_1 + t)$, and we are currently considering the case where t_0 is such that $r_1 \cos(\theta_1 + t) \neq 0$. Since this is precisely the first component of the derivative, we necessarily have that the derivative is nonzero at t_0 . Therefore, the derivative is not identically zero for this case, meaning the velocity is nonzero.

As stated above, the logic for the other cases are precisely the same, so repeating these arguments for the cases $r_1 \sin(\theta_1 + t) \neq 0$ and $r_2 \neq 0$ will yield the result. The velocity is thus everywhere nonzero. ■